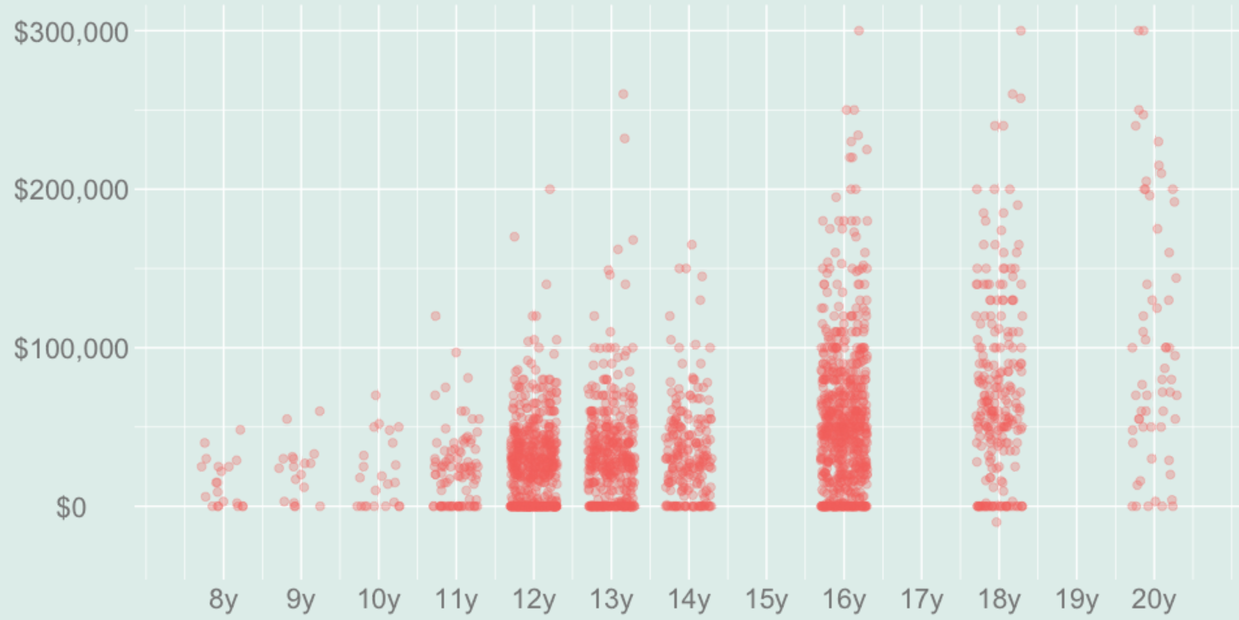
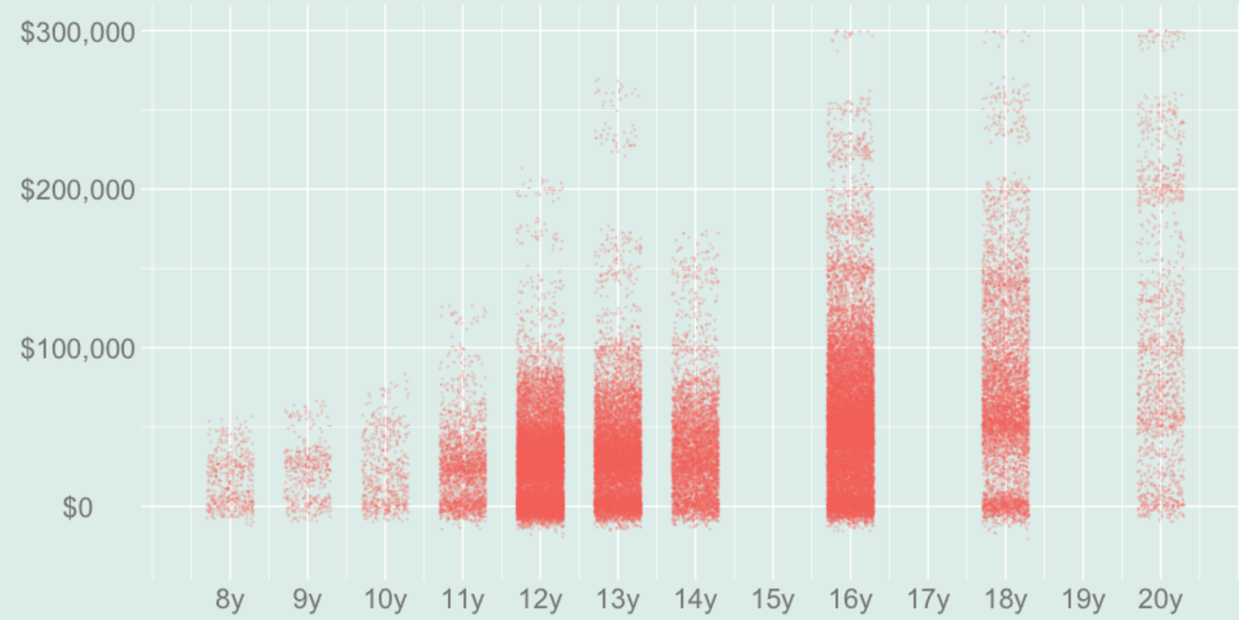
QTM 220 Lecture 11

Inferential Statistics with Non-Binary Covariates: Part 2

Recently





Sample

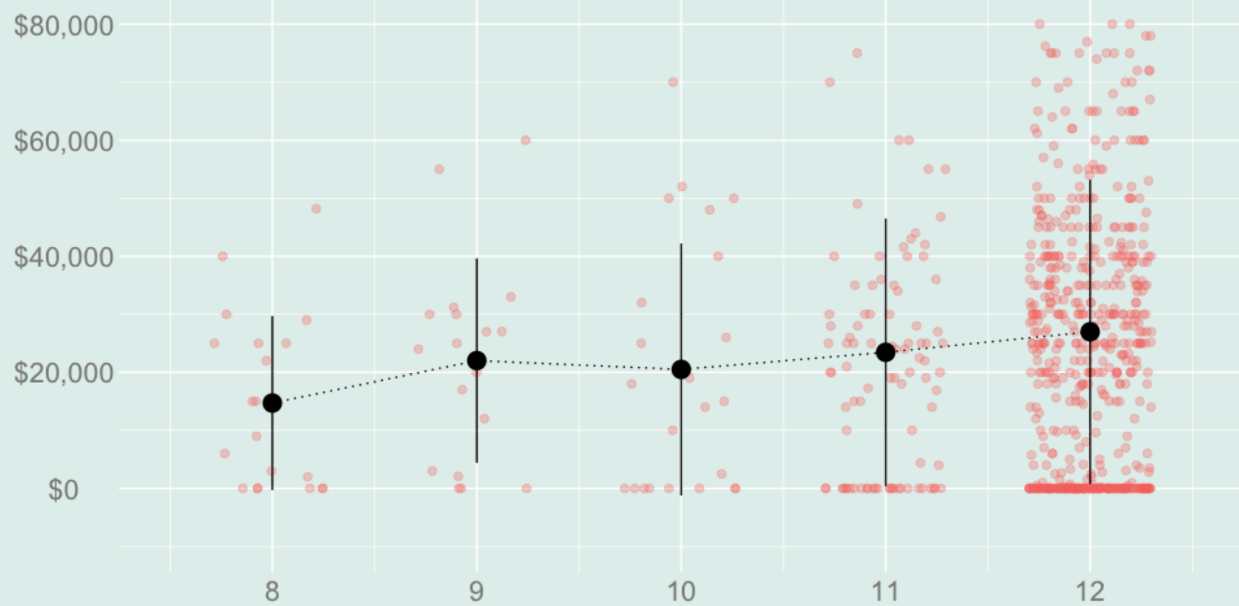
$$
\begin{array}{ccccccc}
X_1 & Y_1 & X_2 & Y_2 & \dots & X_n & Y_n \\
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots
\end{array}
$$

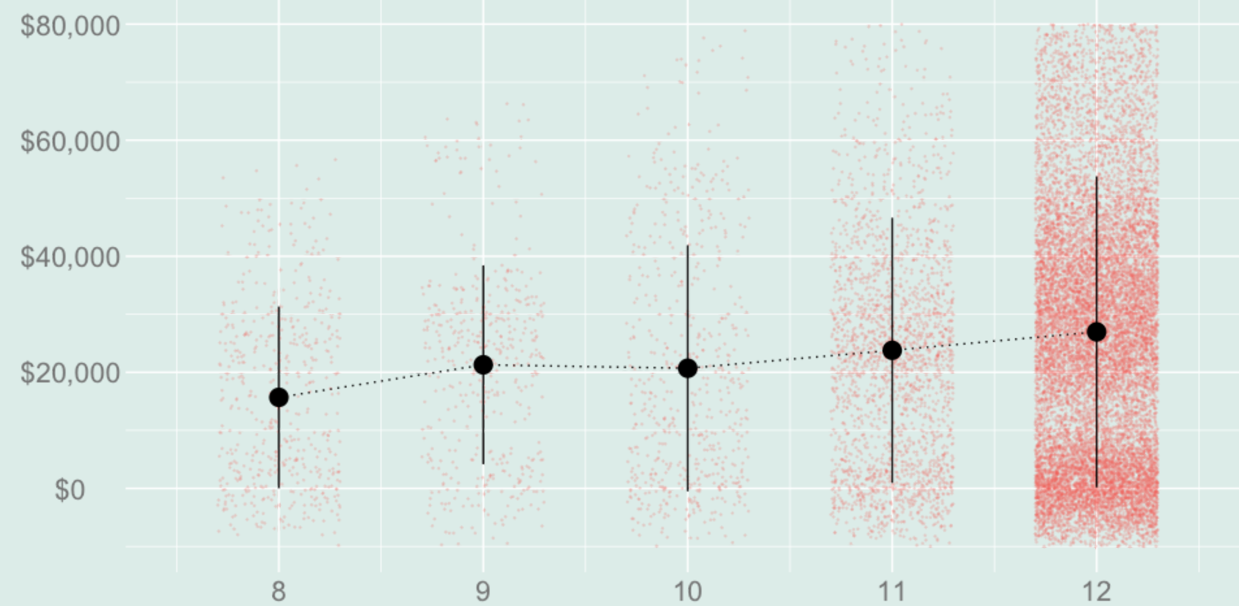
Population

$$
\begin{array}{ccccccc}
x_1 & y_1 & x_2 & y_2 & \dots & x_m & y_m \\
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots
\end{array}
$$

- We've been talking about ways of summarizing data when we have non-binary covariates.
  - → Data that looks like the plot on the left.
- We've talked about how to use those summaries to estimate corresponding *population summaries*.
  - → Assuming the data we've observed is sampled with replacement from that population.
  - → A population that looks like the plot on the right.
- And we've talked about what makes some harder to estimate than others.
  - → i.e., the *variance* of the corresponding sample summaries.

Our Example



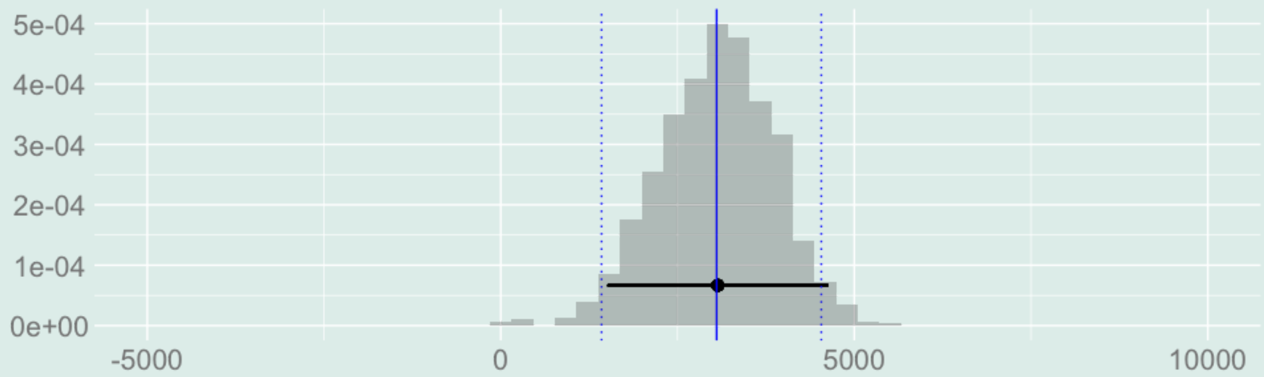


$$\frac{1}{4}\sum_{x=9}^{12}\{\hat\mu(x)-\hat\mu(x-1)\} \quad \text{estimates} \quad \frac{1}{4}\sum_{x=9}^{12}\{\mu(x)-\mu(x-1)\}$$

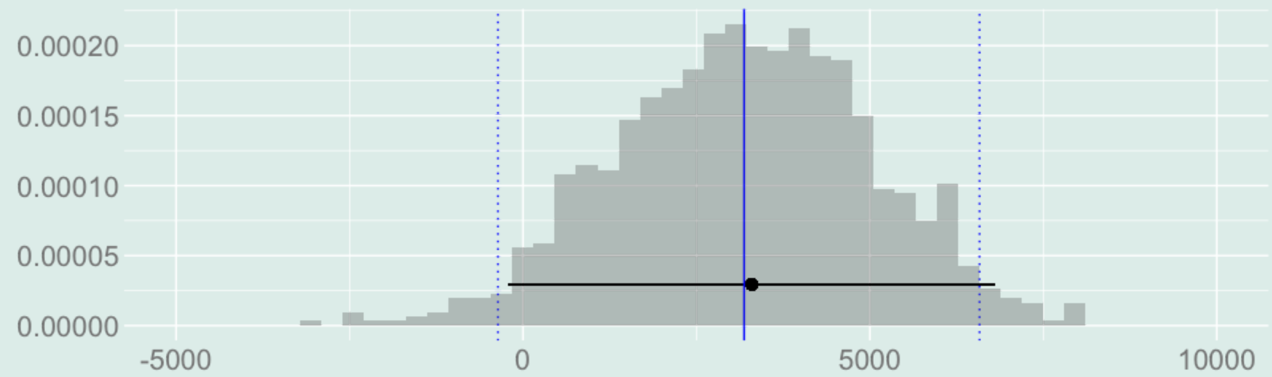
$$\sum_{x=9}^{12} P_x\{\hat\mu(x)-\hat\mu(x-1)\} \quad \text{for} \quad P_x = \frac{N_x}{\sum_{x=9}^{12} N_x} \quad \text{estimates} \quad \sum_{x=9}^{12} p_x\{\mu(x)-\mu(x-1)\} \quad \text{for} \quad p_x = \frac{m_x}{\sum_{x=9}^{12} m_x}$$

- We’ve focused on two main examples. Both summarize the value of a year of high school.
- The first is the mean increment in income associated with *the four years* of high school.
- The second is the mean increment in income associated with *students’ last year* of high school …
  - → where these students are California residents who did not attend college.
    - In our sample summary, we’re talking about *sampled California residents* who didn’t attend college.
    - In our population summary, we’re talking about *all California residents* who didn’t attend college.
  - → This summary speaks to the benefit to these residents of not stopping school a year earlier.

What We See



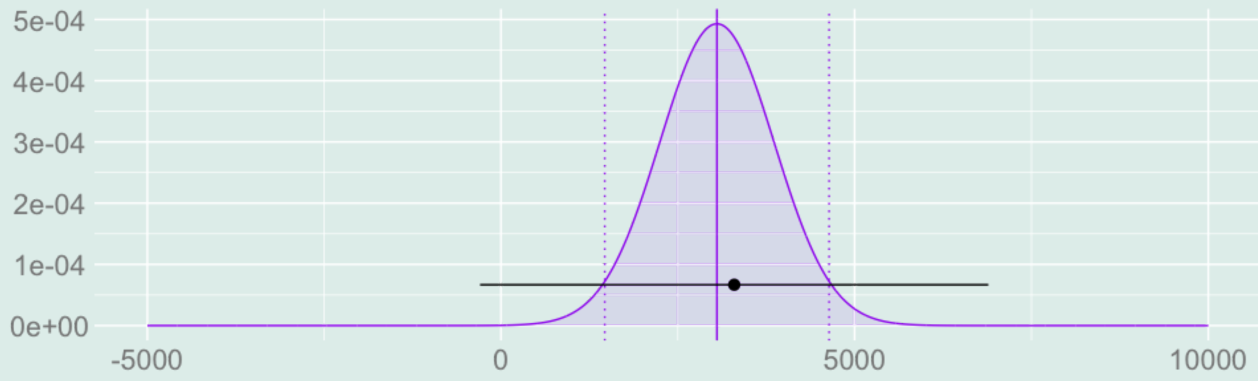
Interval Estimate for the Average over Years



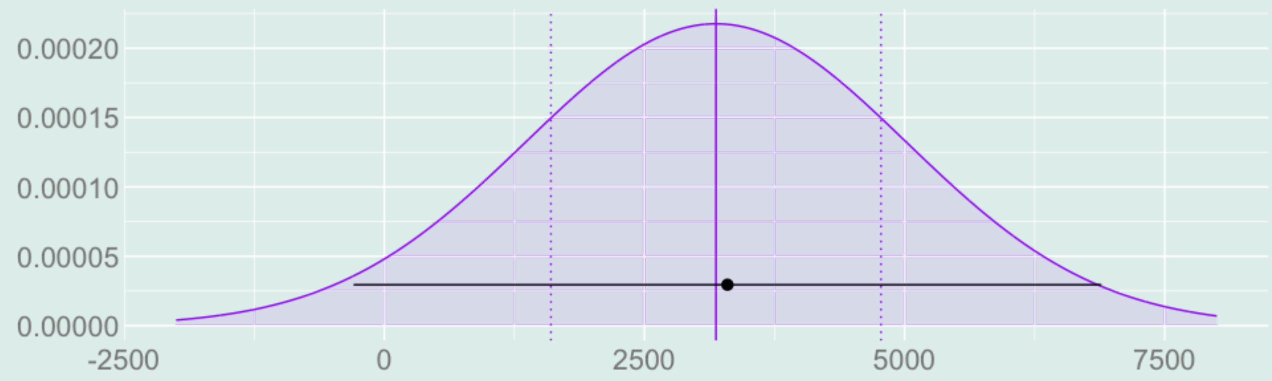
Interval Estimate for the Average over People

- We use *confidence intervals* to characterize our uncertainty about the values of the population summaries.
- And we can calibrate these intervals using the *bootstrap sampling distribution* of our summaries.
- What we see is that we're less certain about the — perhaps more meaningful — average over people.
- We'd like to work out why. And what we can do to get a more precise estimate of what we're interested in.
- e.g. how to use non-representative sampling to get data that's more informative about it.

What'll Be Helpful



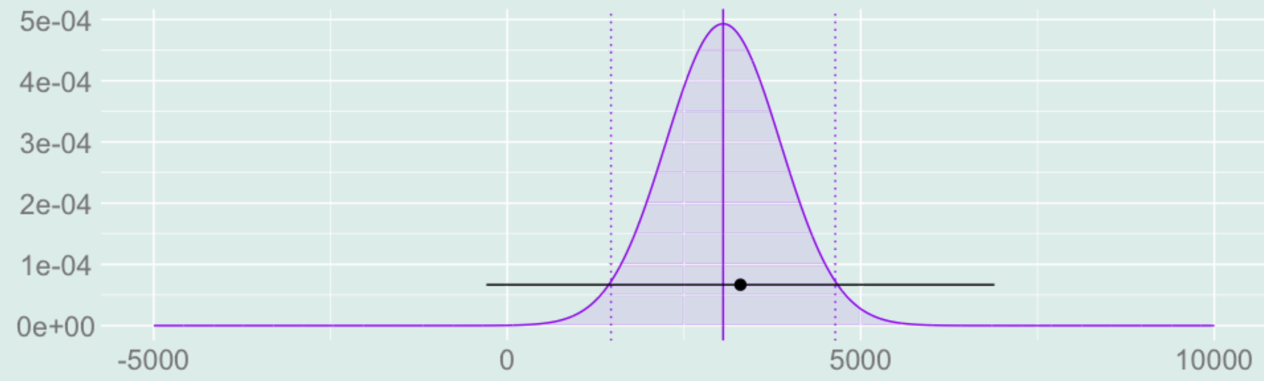
Interval Estimate for the Average over Years



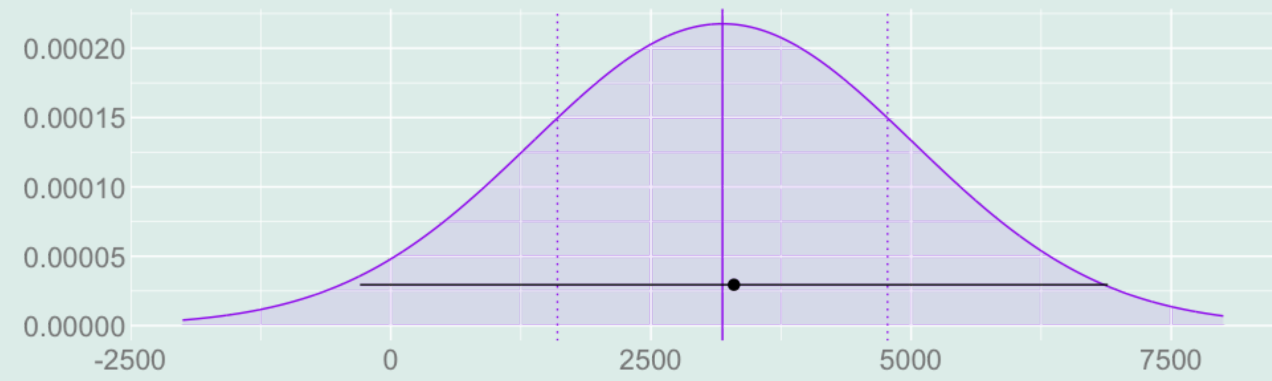
Interval Estimate for the Average over People

- The bootstrap is great if you just want to calibrate an interval. It's automatic and it works (usually).
- But it's not so great if you want to understand what's going on.
  - → These bootstrap histograms we looked at are complicated. Or at least messy.
  - → It'd be nice to boil things down to one or two easy-to-understand numbers.
- Normal approximations do this.
  - → They're simple. If we know the mean and the variance, we know the normal.
  - → And they're usually pretty accurate.
- If we can work out understandable *formulas* for the bias and variance of our summaries …
  - → We can understand what we will and won't be able to estimate *before* collecting our data.
  - → And think about what's driving our uncertainty—e.g. what group(s) we're not sampling enough—and fix it.

An Analogy for Programmers



Interval Estimate for the Average over Years

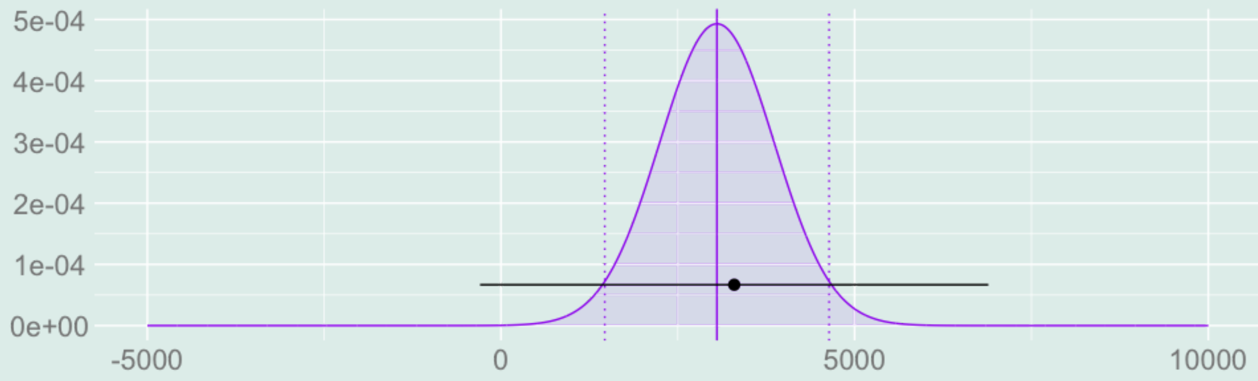


Interval Estimate for the Average over People

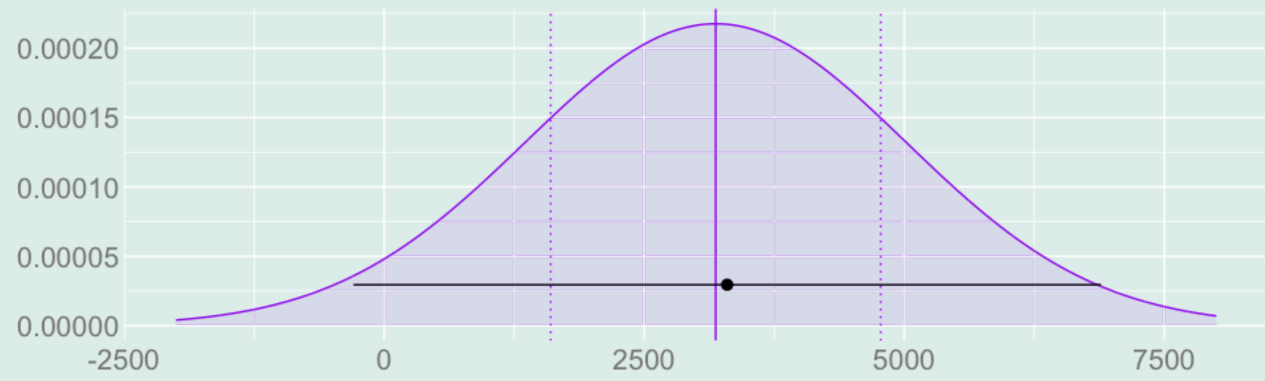
- You can always, of course, just collect more data about everyone.
  - → That'll reduce your uncertainty.
  - → But it's a waste of resources.
- If you're a programmer, think of that like writing all of your code to run as fast as possible.
  - → It's expensive and it's not worth it.
  - → You should focus on the bottlenecks—the part that actually take a meaninful amount of time to run.
- Working out the variance of your summaries is like profiling your code.
  - → It tells you where — in terms of groups you might sample — the bottlenecks are.
  - → And it tells you how much you can expect to gain by sampling more from those groups.

The real problem is that programmers have spent far too much time worrying about efficiency in the wrong places and at the wrong times; premature optimization is the

Where We Are



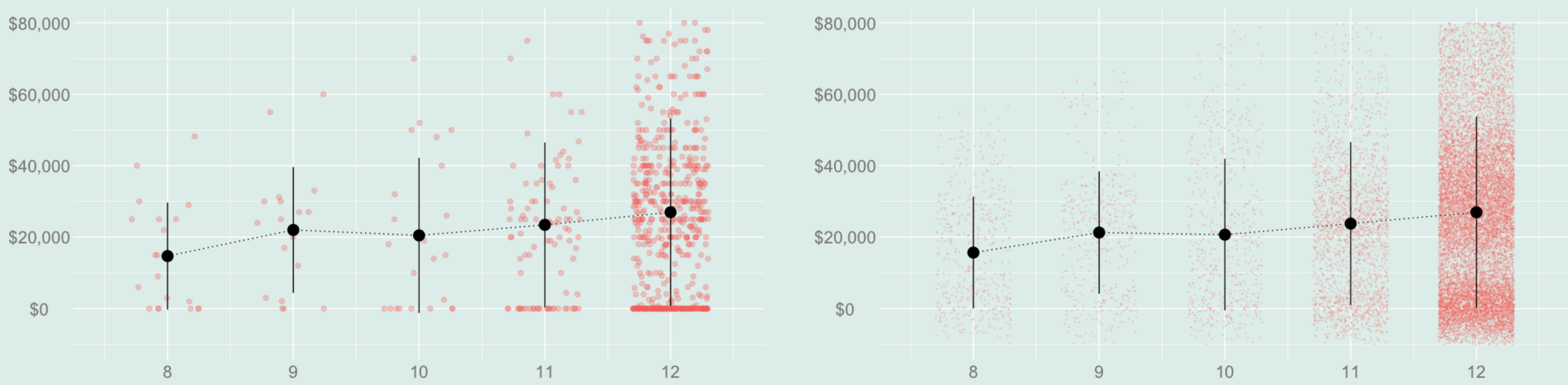
Interval Estimate for the Average over Years



Interval Estimate for the Average over People

- We’ve gotten about half-way to our goal of understanding our two examples.
  - → We’ve worked out a formula for the variance of the average over years.
  - → Today, we’ll do the same for the average over people.

Abstractions



$$\hat{\theta} = \sum_x \hat{\alpha}(x)\hat{\mu}(x) \quad \text{estimates} \quad \theta = \sum_x \hat{\alpha}(x)\hat{\mu}(x)$$

- Rather than tackling our two examples specifically, we'll work in more abstract terms.
- We'll talk about inference for *linear combinations of subsample means* — coefficient-weighted sums of them..
- Working more abstractly is, in a way, easier. We can background some of the details.
- And it's also more useful. It means we can apply our results to other examples, too.
- All we have to do is rewrite our examples in these terms.

Rewriting Our Examples

- Last time, we tackled combinations with *deterministic coefficients*, like the average over years.

$$\frac{1}{4}\sum_{x=9}^{12}\left\{\hat\mu(x)-\hat\mu(x-1)\right\} = \sum_x \alpha(x)\hat\mu(x) \quad \text{for} \quad \alpha(x) = \begin{cases} -\frac{1}{4} & \text{if } x=8 \\ \frac{1}{4} & \text{if } x=12 \\ 0 & \text{otherwise} \end{cases} \qquad \text{estimates}$$

$$\frac{1}{4}\sum_{x=9}^{12}\left\{\mu(x)-\mu(x-1)\right\} = \sum_x \alpha(x)\mu(x) \quad \text{for} \quad \text{the same coefficients}$$

- Today, we'll look at combinations with *random coefficients*, like the average over people.

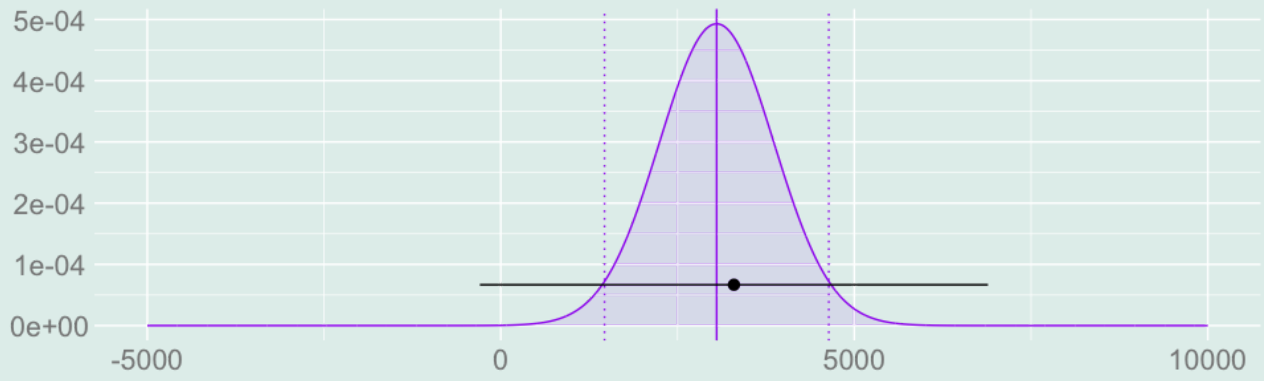
$$\sum_{x=9}^{12} P_x\left\{\hat\mu(x)-\hat\mu(x-1)\right\} = \sum_x \hat\alpha(x)\hat\mu(x) \quad \text{for} \quad \hat\alpha(x) = \begin{cases} P_{12} & \text{if } x=12 \\ P_x - P_{x+1} & \text{if } x\in\{9\ldots 11\} \\ -P_9 & \text{if } x=8 \end{cases} \qquad \text{estimates}$$

$$\sum_{x=9}^{12} p_x\left\{\hat\mu(x)-\hat\mu(x-1)\right\} = \sum_x \alpha(x)\mu(x) \quad \text{for} \quad \alpha(x) = \begin{cases} p_{12} & \text{if } x=12 \\ p_x - p_{x+1} & \text{if } x\in\{9\ldots 11\} \\ -p_9 & \text{if } x=8 \end{cases}$$

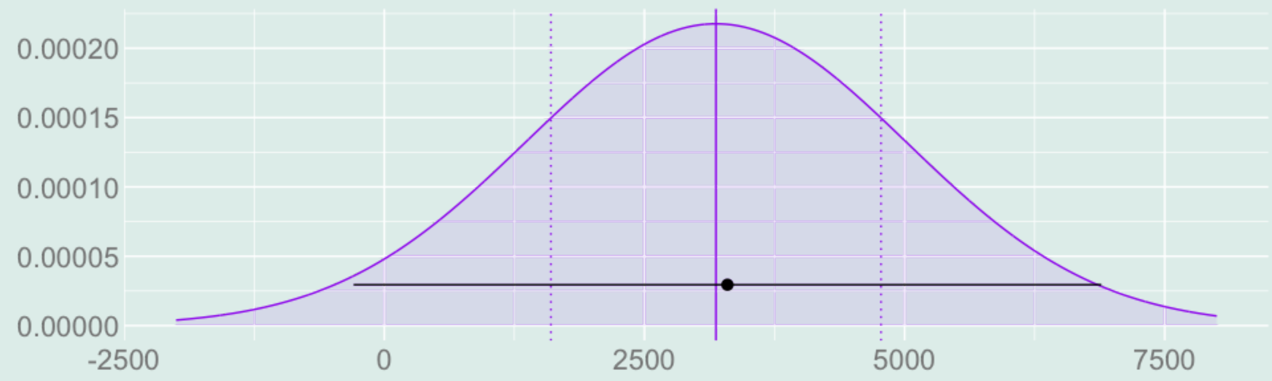
- In these combinations, we're in effect estimating the coefficients $\alpha(x)$.
- This is an additional source of variance. But usually not that much variance. We'll see.

Inference for Linear Combinations of Subsample Means: Random Coefficients

Plan



Interval Estimate for the Average over Years



Interval Estimate for the Average over People

1. We'll show that our estimator is unbiased.

$$\mathrm{E}[\hat{\theta}] = \theta$$

2. We'll calculate — and then estimate — its standard error.

$$\hat{\sigma}_{\hat{\theta}} \approx \sqrt{\mathrm{Var}[\hat{\theta}]}$$

3. And we can use normal approximation to construct a *confidence interval*.

$$\theta \in \left[\hat{\theta} - 1.96\hat{\sigma}_{\hat{\theta}},\ \hat{\theta} + 1.96\hat{\sigma}_{\hat{\theta}}\right]$$

- Working out what contributes to the standard error will help us understand why our intervals are as wide as they are.

Unbiasedness

- Our estimator is unbiased *if the sample coefficients are unbiased estimates of the population coefficients*.
  - → Sample proportions P_x are unbiased for the population proportions p_x.
  - → Differences in Proportions $P_x - P_{x-1}$ are unbiased for the population differences $p_x - p_{x-1}$.
  - → That's enough to handle our example and many other cases.
- The proof is almost the same it was with non-random coefficients.
  - → But now we use linearity of *conditional expectations*.
  - → And *conditional unbiasedness* of the subsample means.

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- The proof is almost the same it was with non-random coefficients.
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  - → And *conditional unbiasedness* of the subsample means.

$$
\begin{aligned}
&\mathrm{E}\left[\sum_x \hat\alpha(x)\hat\mu(x)\right] \\
&= \mathrm{E}\left[\mathrm{E}\left\{\sum_x \hat\alpha(x)\hat\mu(x) \mid X_1 \ldots X_n\right\}\right] && \text{via law of iterated expectations} \\
&= \mathrm{E}\left[\sum_x \hat\alpha(x)\,\mathrm{E}\left\{\hat\mu(x) \mid X_1 \ldots X_n\right\}\right] && \text{via linearity of conditional expectations} \\
&= \mathrm{E}\left[\sum_x \hat\alpha(x)\mu(x)\right] && \text{via conditional unbiasedness of the subsample means} \\
&= \sum_x \mathrm{E}[\hat\alpha(x)]\mu(x) = \sum_x \alpha(x)\mu(x) && \text{via linearity of expectations and unbiasedness of coefficients}
\end{aligned}
$$

Variance Calculation Step 1: Total Variance

- We'll use the law of total variance to break the variance of our estimator into two parts.
- One of these parts — $\mathrm{Var}\,\mathrm{E}[\ldots]$ part — has always been zero before.
  - → It isn't now. This part is how the *randomness of the coefficients* contributes.
  - → It was zero last time because the coefficients were constants.
- What we'll get for the remaining $\mathrm{E}[\mathrm{Var}[\ldots]]$ part is almost exactly what we did before.
  - → We *sum* the conditional variances of the individual terms, then take the expectation of the result.
  - → All that changes is that, the coefficients being random, we can't pull them out of the expectation

$$
\begin{aligned}
\mathrm{Var}\left[\sum_x \alpha(x)\hat\mu(x)\right] &= \mathrm{E}\left[\mathrm{Var}\left\{\sum_x \hat\alpha(x)\hat\mu(x) \mid X_1\ldots X_n\right\}\right] &&+ \quad \mathrm{Var}\left[\mathrm{E}\left\{\sum_x \hat\alpha(x)\hat\mu(x) \mid X_1\ldots X_n\right\}\right] \\
&= \mathrm{E}\left[\mathrm{Var}\left\{\sum_x \hat\alpha(x)\hat\mu(x) \mid X_1\ldots X_n\right\}\right] &&+ \quad \mathrm{Var}\left[\sum_x \hat\alpha(x)\,\mathrm{E}\left\{\hat\mu(x) \mid X_1\ldots X_n\right\}\right] \\
&= \mathrm{E}\left[\mathrm{Var}\left\{\sum_x \hat\alpha(x)\hat\mu(x) \mid X_1\ldots X_n\right\}\right] &&+ \quad \mathrm{Var}\left[\sum_x \hat\alpha(x)\mu(x)\right] \\
&= \sum_x \mathrm{E}\left[\hat\alpha(x)^2\,\mathrm{Var}\left\{\hat\mu(x) \mid X_1\ldots X_n\right\}\right] &&+ \quad \sum_x\sum_{x'} \mu(x)\mu(x')\,\mathrm{Cov}\left[\hat\alpha(x),\ \hat\alpha(x')\right] \\
&= \sum_x \sigma^2(x) \times \mathrm{E}\left[\frac{\hat\alpha(x)^2}{N_x}\right] &&+ \quad \sum_x\sum_{x'} \mu(x)\mu(x')\,\mathrm{Cov}\left[\hat\alpha(x),\ \hat\alpha(x')\right]
\end{aligned}
$$

Variance Calculation Step 2a: $\mathbf{E}[\mathbf{Var}[\ldots]]$

$$
\begin{aligned}
\mathrm{E}\left[\mathrm{Var}\left\{\sum_x \hat\alpha(x)\hat\mu(x) \mid X_1\ldots X_n\right\}\right] &= \mathrm{E}\left[\mathrm{E}\left\{\left(\sum_x \hat\alpha(x)\hat\mu(x) - \mathrm{E}\left[\sum_x \alpha(x)\hat\mu(x) \mid X_1\ldots X_n\right]\right)^2 \mid X_1\ldots X_n\right\}\right] \\
&= \mathrm{E}\left[\mathrm{E}\left\{\left(\sum_x \hat\alpha(x)\hat\mu(x) - \alpha(x)\,\mathrm{E}\left[\hat\mu(x) \mid X_1\ldots X_n\right]\right)^2 \mid X_1\ldots X_n\right\}\right] \\
&= \mathrm{E}\left[\mathrm{E}\left\{\left(\sum_x \hat\alpha(x)\left\{\hat\mu(x) - \mathrm{E}\left[\hat\mu(x) \mid X_1\ldots X_n\right]\right\}\right)^2 \mid X_1\ldots X_n\right\}\right] \\
&= \mathrm{E}\left[\mathrm{E}\left\{\left(\sum_x \hat\alpha(x)Z_x\right)^2 \mid X_1\ldots X_n\right\}\right] \quad \text{for} \quad Z_x = \hat\mu(x) - \mathrm{E}\left[\hat\mu(x) \mid X_1\ldots X_n\right]
\end{aligned}
$$

- Here Z_x is a *conditionally centered* version of our subsample mean $\hat\mu(x)$.
  - → The *conditional variance* of $\hat\mu(x)$ is the conditional expectation of its square Z_x^2
  - → And the conditional expectation of *products* $Z_x Z_{x'}$ for different subsamples is zero.

$$
\begin{aligned}
\mathrm{E}[Z_x^2 \mid X_1\ldots X_n] &= \mathrm{Var}[\hat\mu(x) \mid X_1\ldots X_n] \\
\mathrm{E}[Z_x Z_{x'} \mid X_1\ldots X_n] &= 0 \quad \text{for} \quad x \neq x'
\end{aligned}
$$

Variance Calculation Step 2b: $\mathbf{E}[\mathbf{Var}[\ldots]]$ and FOIL

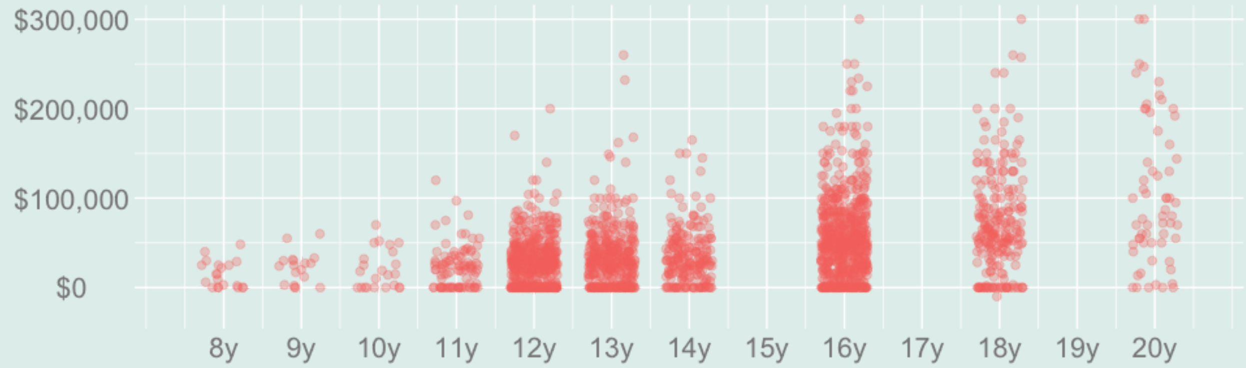
- To make progress from here, we have to square out our sum. We get one term for each combination of xs.

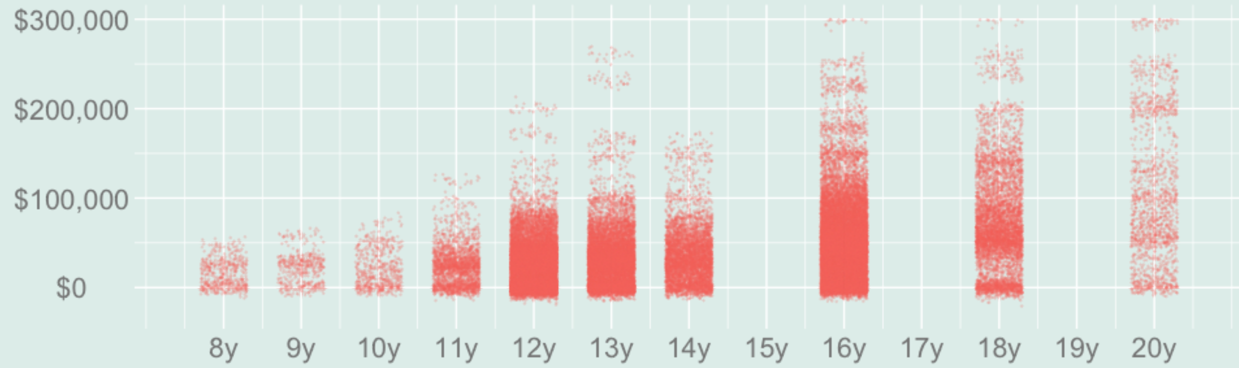
$$\left(\sum_x \hat\alpha_x Z_x\right)^2 = \sum_x \sum_{x'} \hat\alpha_x Z_x \; \hat\alpha_{x'} Z_{x'}$$

- Making this substitution, we can boil things down to something we've calculated before.

$$\begin{aligned}
\mathrm{E}\left[\mathrm{Var}\left\{\sum_x \hat\alpha(x)\hat\mu(x) \mid X_1 \ldots X_n\right\}\right] &= \mathrm{E}\left[\mathrm{E}\left\{\left(\sum_x \hat\alpha(x) Z_x\right)^2 \mid X_1 \ldots X_n\right\}\right] \\
&= \mathrm{E}\left[\mathrm{E}\left\{\sum_x \sum_{x'} \hat\alpha(x)\hat\alpha(x') Z_x Z_{x'} \mid X_1 \ldots X_n\right\}\right] \\
&= \mathrm{E}\left[\sum_x \sum_{x'} \hat\alpha(x)\hat\alpha(x')\,\mathrm{E}\left\{Z_x Z_{x'} \mid X_1 \ldots X_n\right\}\right] \\
&= \mathrm{E}\left[\sum_x \sum_{x'} \hat\alpha(x)\hat\alpha(x') \begin{cases} \mathrm{Var}\left\{\hat\mu(x) \mid X_1 \ldots X_n\right\} & \text{if } x = x' \\ 0 & \text{otherwise} \end{cases}\right] \\
&= \mathrm{E}\left[\sum_x \hat\alpha(x)^2\,\mathrm{Var}\left\{\hat\mu(x) \mid X_1 \ldots X_n\right\}\right]
\end{aligned}$$

Conclusion for *This Part* of the Variance

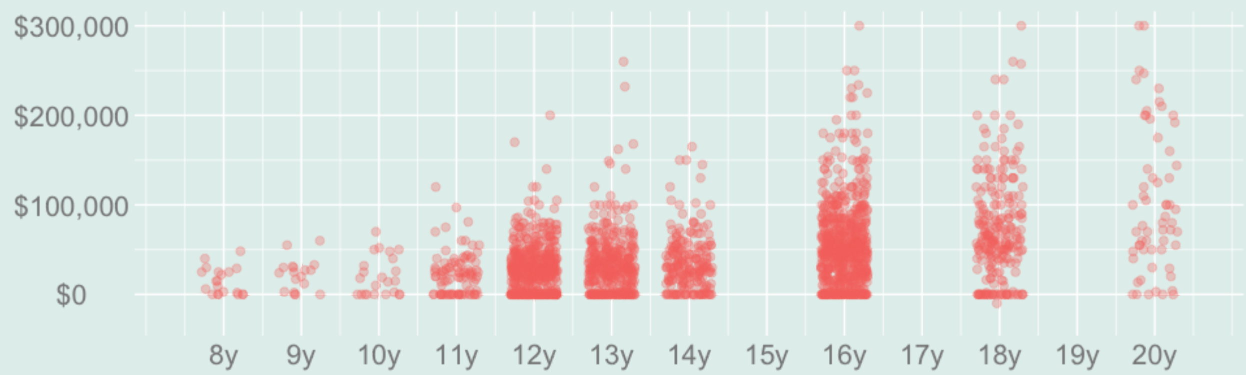


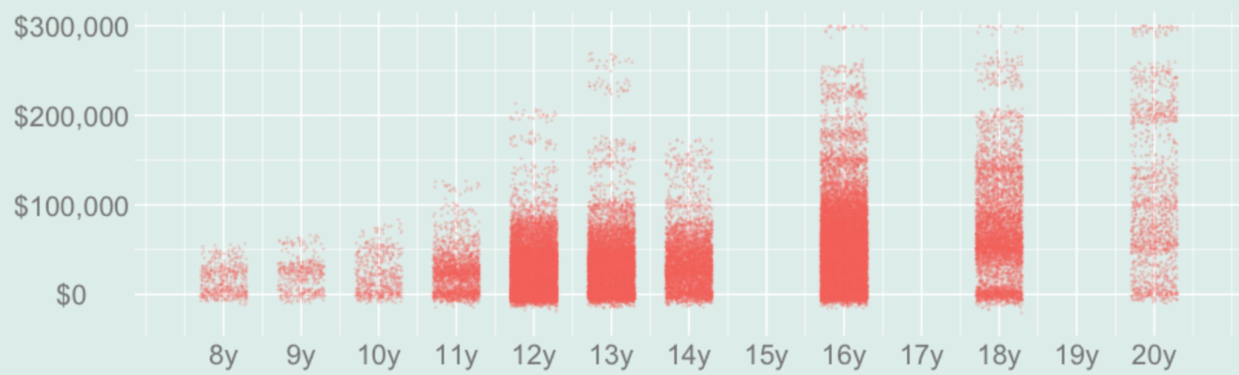


- In the end, this part of the variance is essentially what it was last time.
- It's a linear combination of the conditional variances of the subsample means.
- We worked out how to calculate those before the midterm.

$$
\begin{aligned}
\mathrm{E}\left[\mathrm{Var}\left\{\sum_x \hat\alpha(x)\hat\mu(x) \mid X_1 \dots X_n\right\}\right] &= \mathrm{E}\left[\sum_x \hat\alpha(x)^2\, \mathrm{Var}\left\{\hat\mu(x) \mid X_1 \dots X_n\right\}\right] \\
&= \mathrm{E}\left[\sum_x \hat\alpha(x)^2 \frac{\sigma^2(x)}{N_x}\right] \quad \text{where} \quad \sigma^2(x) = \mathrm{Var}[Y_i \mid X_i = x] \\
&= \sum_x \sigma^2(x) \times \mathrm{E}\left[\frac{\hat\alpha(x)^2}{N_x}\right]
\end{aligned}
$$

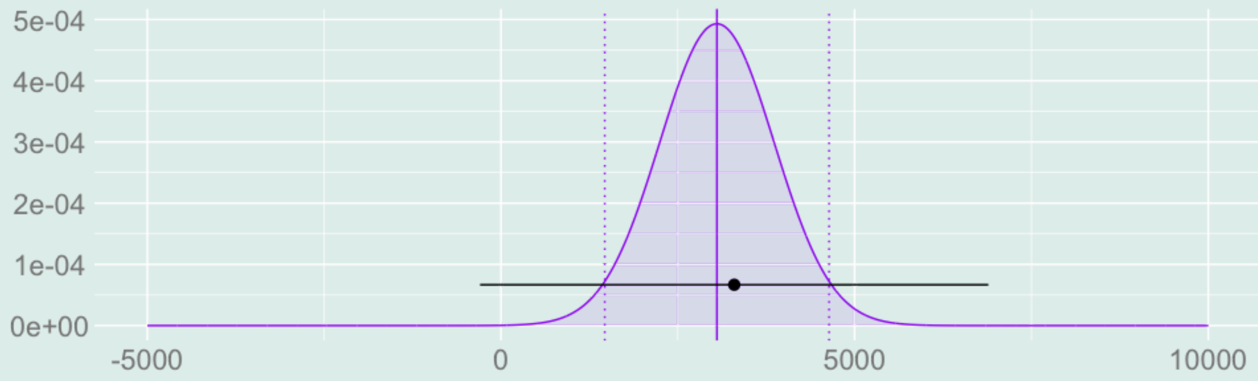
Estimating This Part



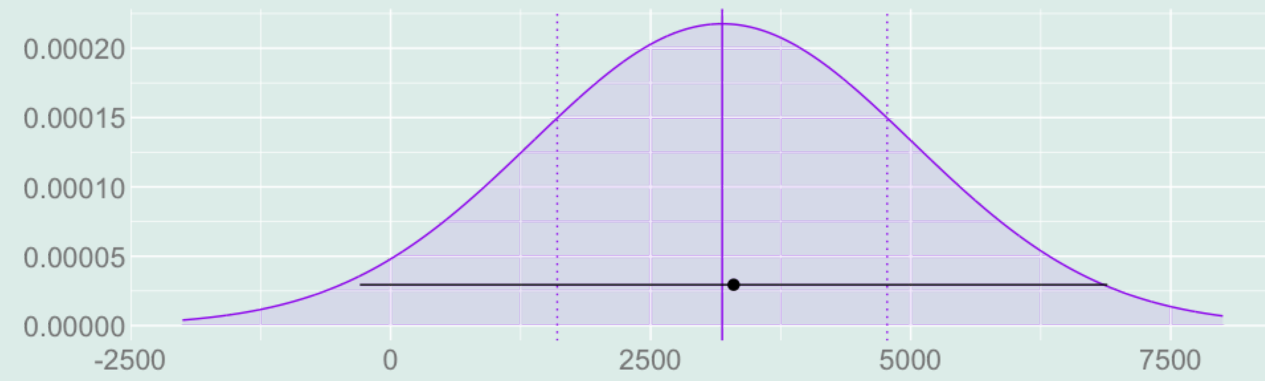


$$\mathrm{E}\left[\mathrm{Var}\left\{\sum_x \hat{\alpha}(x)\hat{\mu}(x) \mid X_1 \ldots X_n\right\}\right] = \sum_x \sigma^2(x) \times \mathrm{E}\left[\frac{\hat{\alpha}(x)^2}{N_x}\right] \quad \text{where} \quad \sigma^2(x) = \mathrm{Var}[Y_i \mid X_i = x]$$

A Pause to Consider the Implications



Interval Estimate for the Average over Years



Interval Estimate for the Average over People

$$\mathrm{Var}\left[\sum_x \hat\alpha(x)\hat\mu(x)\right] \ge \mathrm{E}\left[\mathrm{Var}\left\{\sum_x \hat\alpha(x)\hat\mu(x) \mid X_1 \ldots X_n\right\}\right] = \sum_x \sigma^2(x) \times \mathrm{E}\left[\frac{\hat\alpha(x)^2}{N_x}\right]$$

- Let’s think about what this means for our two examples.

$$\hat\theta_{\text{years}} = \frac{1}{4}\sum_{x=9}^{12}\{\hat\mu(x) - \hat\mu(x-1)\} = \sum_x \alpha(x)\hat\mu(x) \quad \text{for} \quad \alpha(x) = \begin{cases} \frac{1}{4} & \text{if } x = 12 \\ -\frac{1}{4} & \text{if } x = 8 \\ 0 & \text{otherwise} \end{cases}$$

$$\hat\theta_{\text{people}} = \sum_{x=9}^{12} P_x\{\hat\mu(x) - \hat\mu(x-1)\} = \sum_x \hat\alpha(x)\hat\mu(x) \quad \text{for} \quad \hat\alpha(x) = \begin{cases} P_{12} & \text{if } x = 12 \\ P_x - P_{x+1} & \text{if } x \in \{9 \ldots 11\} \\ -P_9 & \text{if } x = 8 \end{cases}$$

- We’ve seen that our estimate of the average-over-years summary of the value of a year of high school has large variance.
- The problem was that it paired a relatively large weight—$\alpha(x) = -1/4$—with a very small sample size $N_8 = 20$.
- And we know the average-over-people summary has an even larger variance. Let’s compare.

A Comparison

	x	8	9	10	11	12	$\sum$
	N_x	20	18	23	93	584	738
	$\hat{\sigma}^2(x)$	$225M$	$310M$	$472M$	$532M$	$687M$	
for $\hat{\theta}_{\text{people}}$	$\hat{\alpha}^2(x)$	0.00	0.00	0.01	0.44	0.63	
	$\frac{\hat{\sigma}^2(x)\alpha^2(x)}{N_x}$	$7K$	790	$185K$	$2.5M$	$737K$	$3.5M$
for $\hat{\theta}_{\text{years}}$	$\alpha^2(x)$	0.06	0.00	0.00	0.00	0.06	
	$\frac{\hat{\sigma}^2(x)\alpha^2(x)}{N_x}$	$703K$	0	0	0	$74K$	$777K$

- For both summaries, *most* of the variance comes from one term—from the variance of one subsample mean.
- For the average over years, it's the $x = 8$ subsample, with a sample size of $N_x = 20$ and a coefficient of $-1/4$.
- For the average over people, it's the $x = 11$ subsample, with …
  - → a sample size $N_x = 93$ that's roughly 4x bigger.
  - → but a coefficient of -0.665 that's roughly 3x bigger, too.
- Variance grows with the *square* of the coefficient divided by the sample size …
- So this suggests variance would be roughly $3^2/4 \approx 2.25x$ bigger.
- There's actually a bit more to it—$\sigma^2(11) \approx 2 \times \sigma^2(8)$ — so it's really 5x bigger.

What About The Other Part?

$$\mathrm{Var}\left[\sum_x \alpha(x)\hat{\mu}(x)\right] = \mathrm{E}\left[\mathrm{Var}\left\{\sum_x \hat{\alpha}(x)\hat{\mu}(x) \mid X_1 \ldots X_n\right\}\right] \quad + \quad \mathrm{Var}\left[\mathrm{E}\left\{\sum_x \hat{\alpha}(x)\hat{\mu}(x) \mid X_1 \ldots X_n\right\}\right]$$

- What we've been thinking about so far is the first part of the variance.
  - → That's zero when the coefficients are constants, but not when they're random.
  - → Because this part was *larger* for the summary with random coefficients, that implied its variance would be larger.
  - → But are we missing something? Are we underestimating how much larger it'll be by ignoring the second part?
- Basically no.
  - → We can estimate the variance of our summary—the whole thing—by the variance of bootstrap sampling distribution. It's 3.4M.
  - → Our estimate for the first part only — based on our table — was 3.5M.
  - → That suggests that the second part should be roughly $3.4M - 3.5M \approx -100K$.
  - → That's impossible—the part is the variance of something, so it can't be negative—but it suggests it's not very big.
  - → Let's see why.

Variance Calculation Step 3: The $\mathbf{Var}[\mathbf{E}[\ldots]]$ Part.

- Now we'll calculate the new part — the $\mathbf{Var}[\mathrm{E}[\ldots]]$ part.
- This **additional variance** is what we pay for using *estimates* of the population coefficients $\alpha(x)$.
- We're going to use a formula for the variance of a linear combination of random variables that you may have seen before.

$$\mathrm{Var}\left[\sum_x a_x Z_x\right] = \sum_x \sum_{x'} a_x a'_x \operatorname{Cov}(Z_x,\ Z_y) \quad \text{for} \quad \operatorname{Cov}(Z_x,\ Z_{x'}) = \mathrm{E}[\{Z_x - \mathrm{E}[Z_x]\}\{Z_{x'} - \mathrm{E}[Z_{x'}]\}]$$

- You may have seen a special case instead, where the Z_x are *independent*—i.e. where $\operatorname{Cov}(Z_x,\ Z_{x'}) = 0$ for $x \neq x'$.
- This gives us the simple statement *the variance of a sum is the sum of the variances*—but that's true only when the Z_x are independent.
- And ours, generally, won't be. $\hat\alpha(12) = P_{12}$ and $\hat\alpha(11) = P_{11} - P_{12}$ are *not* independent.
- This general version is easy enough to prove, but we'll skip that. Here's what it tells us.

$$\mathrm{Var}\left[\mathrm{E}\sum_x \hat\alpha(x)\hat\mu(x) \mid X_1 \ldots X_n\right] = \mathrm{Var}\left[\sum_x \mu(x)\hat\alpha(x)\right] = \sum_x \sum_{x'} \mu(x)\mu(x') \operatorname{Cov}[\hat\alpha(x),\ \hat\alpha(x')]$$

Variance Calculation Step 4?

$$\mathrm{Var}\left[\mathrm{E}\sum_x \hat{\alpha}(x)\hat{\mu}(x) \mid X_1 \ldots X_n\right] = \mathrm{Var}\left[\sum_x \mu(x)\hat{\alpha}(x)\right] = \sum_x \sum_{x'} \mu(x)\mu(x')\,\mathrm{Cov}\left[\hat{\alpha}(x),\ \hat{\alpha}(x')\right]$$

- These covariances are easy enough to calculate, but doing all of them is a bit tedious.
- We'll just do enough to get a handle on the size of this additional variance.
- And we'll use a simple *upper bound* on the covariances to do it.
- The covariance of two random variables is no larger than *the product of their standard deviations*.

$$\mathrm{Cov}\left[\hat{\alpha}(x),\ \hat{\alpha}(x')\right] \le \sqrt{\mathrm{Var}[\hat{\alpha}(x)]} \times \sqrt{\mathrm{Var}[\hat{\alpha}(x')]}$$

- Now let's calculate one of these standard deviations. We'll stick with a simple one: $\hat{\alpha}(12) = P_{12}$.
- This is roughly the square root of $p_x(1-p_x)/\sum_x N_x$ for $x = 12$
  - → i.e., the standard deviation of an average of Bernoulli random variables with expectation p_{12}.
- What's important is that *the denominator here is the sum of the sample sizes*.
  - → It's not small like the denominators in the other part.
  - → As a result, this additional part of the variance is *much smaller* than the other one.

Variance Calculation Step 4: Never Worth It?

$$\mathrm{Var}\left[\mathrm{E}\sum_x \hat\alpha(x)\hat\mu(x) \mid X_1 \ldots X_n\right] = \mathrm{Var}\left[\sum_x \mu(x)\hat\alpha(x)\right] = \sum_x \sum_{x'} \mu(x)\mu(x')\,\mathrm{Cov}\left[\hat\alpha(x),\ \hat\alpha(x')\right]$$

- I've just argued that the second part of the variance is small.
- That had to do with the weights $\hat\alpha(x)$ not being all that variable.
- This is the case for this example, but it's not always the case.
- We'll see one or two later in the semester

Activity: Alternative Sampling Schemes

	x	8	9	10	11	12	$\sum$
	N_x	?	?	?	?	?	1000
	$\hat{\sigma}^2(x)$	$1M$	$1M$	$1M$	$1M$	$1M$	
for $\hat{\theta}_{\text{people}}$	$\hat{\alpha}^2(x)$	0.00	0.00	0.01	0.44	0.63	
	$\frac{\hat{\sigma}^2(x)\alpha^2(x)}{N_x}$	?	?	?	?	?	
for $\hat{\theta}_{\text{years}}$	$\alpha^2(x)$	0.06	0.00	0.00	0.00	0.06	
	$\frac{\hat{\sigma}^2(x)\alpha^2(x)}{N_x}$	?	?	?	?	?	

- Suppose we're considered non-representative sampling. In effect, we can decide our subsample sizes N_x.
- So for each summary, propose a way of allocating 1000 observations to our 5 subsamples to reduce variance.
  - → i.e., choose $N_8 \ldots N_{12}$, ensuring they sum to 1000.
- Then swap proposals with a neighbor and …
  - → Calculate the variance of each summary using the allocation they propose to reduce its variance.
  - → Compare the `optimized' variances you've calculated for these two summaries. Which is smaller?
  - → Do you think that reflects the difficulty of estimating one summary vs. the other?
  - → Or is your neighbor's proposal just not very good? Did they do better with yours?
- For simplicity …
  - → Assume the income variance within each subsample is the same—$\sigma^2(x) = 1M$ for all x.
  - → Ignore the second part of the variance—the $\mathbf{Var}[\mathbf{E}[\ldots]]$ part.